Ashim Shrestha

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Professional Summary

AI/ML engineer with hands-on experience across computer vision, reinforcement learning, and IoT-driven automation, paired with full-stack web development experience as a startup co-founder. Proficient in PyTorch, with a track record of building and shipping end-to-end systems spanning embedded sensing, real-time inference, and production web applications. STEAM Competition Grant recipient and President of the Kathmandu University AI Club.

Education

Kathmandu University, Kavre, Dhulikhel, Nepal *Expected Graduation: 2027*
Bachelor of Technology in Artificial Intelligence
Relevant Coursework: Deep Learning, Pattern Recognition , Advanced Machine Learning , Large Language Models, Graph Neural Networks, GIS & Remote Sensing , Information Retreival

Technical Skills

Programming Languages:	Python, C++, C, JavaScript, SQL, Julia
Machine Learning & Deep Learning:	PyTorch (Proficient), TensorFlow/Keras, Scikit-learn, ONNX, Hugging Face, CNNs, Transformers, Meta-Learning, Vision Transformers
Reinforcement Learning:	Gymnasium, Stable-Baselines3, PPO, DQN
Computer Vision:	OpenCV, YOLOv8, OCR, Object Detection, Face Recognition, Image Classification, Prototypical Networks
Web & Full-Stack Development:	MERN Stack (MongoDB, Express.js, React, Node.js), FastAPI, Flutter, WebRTC, RabbitMQ
Embedded Systems & IoT:	ESP32, Raspberry Pi, Arduino, MQTT, Sensor Fusion , PCB Design
Tools & Platforms:	Git, Docker, Linux, QGIS

Professional Experience

Sichu Nepal - Co-Founder & Chief Technology Officer *Start Year – Present*

- Co-founded an e-commerce startup selling anime, Marvel, and pop-culture merchandise, and led all technical development.
- Architected and built a full-stack MERN (MongoDB, Express.js, React, Node.js) web platform covering product catalog, cart, checkout, and order management.
- Directed technical strategy and infrastructure decisions as CTO, balancing product growth with engineering scalability.

Research & Development Projects

LipiNet: A Novel Inductive-Bias Approach for Multi-Script Handwritten Text Recognition

- Proposed a handwritten text recognition (HTR) architecture that embeds structural inductive biases to improve recognition across multiple character sets.
- Benchmarked performance against conventional OCR/HTR baselines to validate generalization gains.
- Independent research-focused project exploring architecture design for low-resource script recognition.

Multi-Script OCR for Ancient Nepali Manuscripts using Meta-Learning

- Designed a parameter-efficient meta-learning architecture for recognizing seven historical Nepali scripts.
- Introduced architectural refinements that improved accuracy while reducing model complexity.
- Demonstrated cross-script generalization under limited labeled data per script, supporting digital preservation of cultural heritage documents.

Hydroponics AI (Smart Soilless Farming) - RL, FastAPI, MQTT, Flutter, IOT, RabbitMQ

2025

- Co-developed a closed-loop IoT system using ESP3 with a FastAPI backend for MQTT telemetry and automated actuator control.
- Implemented a Proximal Policy Optimization (PPO) reinforcement learning agent for autonomous nutrient dosing and statistical anomaly detection for sensor fault handling.
- Deployed custom algorithmic pipelines for canopy and disease analysis, with live monitoring through a Flutter application.

Open Set Facial Recognition & Surveillance – PyTorch, TensorFlow, FastAPI, WebRTC, Electron, MongoDB

2024

- Built an open-set surveillance system using convolutional neural networks, N-shot learning, and KNN to identify registered individuals and flag unknown faces in real time.
- Integrated WebRTC for low-latency, real-time video streaming between camera nodes and an Electron-based operator dashboard, enabling live monitoring and instant alerts.
- Used MongoDB for large-scale storage of facial embeddings and surveillance logs.

Reaco - AI-Powered Face Detection and Photo Sorting – YOLOv8, InceptionV3, Siamese Network, KNN

2024

- Built an automated face detection and clustering pipeline using YOLOv8 face detection, Dlib alignment, and an InceptionV3 Siamese embedding model with KNN-based grouping.
- Developed a Flutter/Dart front end backed by FastAPI, MongoDB, and Docker for image upload, face registration, and sorted photo retrieval across large collections.

Landslide Susceptibility Mapping - Nepal - QGIS, Python, rasterio, Scikit-learn

2025

- Co-developed machine learning-based landslide susceptibility maps for Nepal by fusing Landsat composites, Digital Elevation Model derivatives, and geo-features into a supervised classification pipeline.

Parikar (Nepali Food Classifier) – TensorFlow/Keras, YOLOv8, InceptionV3, VGG16, Flutter

2023

- Built a Nepali food classification mobile application using transfer learning in 12 food categories.
- Converted trained models to TensorFlow Lite and deployed inference in a cross-platform Flutter application, achieving 90%+ accuracy.

Leadership & Activities

President, Kathmandu University AI Club (KUAIC) *Start Year – Present*

- Lead technical and organizational initiatives promoting AI education across the university.
- Organize workshops, competitions, hackathons, and research events; coordinate collaborations between students, faculty, and industry partners.
- Oversee club operations including the AI Conclave, a flagship national-level event.

Awards & Achievements

- **STEAM Competition Grant Recipient**